# 8. Engineered a macOS SSL Bypass & Vercel Fix

## Overview

During the deployment of the new architecture, we hit a critical local development blocker: macOS's strict HTTPS context was preventing Python's urllib from securely downloading the .joblib models from GitHub. Additionally, we needed to route the correct GitHub branches to Vercel for production hosting.

## Code Changes

- **Terminal Bypass:** Engineered a dynamic Python terminal wrapper (python3 -c "import ssl, runpy; ssl._create_default_https_context = ssl._create_unverified_context; runpy.run_path('app.py', run_name='__main__')") to globally disable strict certificate checks for that specific process without needing to hardcode insecure bypasses into the actual app.py production files.
- **Deployment:** Configured Vercel's production branch settings to sync directly with the main branch of the original upstream repository.

## Future Roadmap

To permanently eliminate local environment inconsistencies like macOS SSL failures, the next step is to containerize the entire Flask backend using Docker. This will ensure that the server, dependencies, and SSL certificates operate identically whether running on a local Mac, a Windows machine, or a cloud server.